

Fluid Flow in Pipe Systems

Pre-Lab Plan
Unit Operations Lab 6
Date: 11/14/19

Group 4, Thursday, PM section
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1. Experimental Objective

The purpose of this experiment is to study the correlation between theoretical and experimental data for a fluid flowing through system of pipes. More specifically the goal is to determine the friction factor, f , loss coefficient, K , and flow coefficient, C_f , at various parts of the piping system containing water. To do this we will be measuring pressure drop, flow rate, and various other measurements from the piping system.

2. Experimental

2.1 Apparatus

The apparatus setup for this experiment includes pipes of various sizes, a pump, and valves as well as devices to measure the parameters necessary for correlating the experimental data to theoretical data. The following figure (Figure 1) and table (Table 1) summarize the setup for this lab.



Figure 1: Apparatus setup for Fluid Flow in Pipe Systems experiment

Table 1: Hampden H-6920 Pipe Friction Demonstrator Component Description

Number	Component Name	Description
1	Orifice plate	Measures rate of fluid flow by recording pressure drop across hole
2	Pump	Pumps water into process
3	Reservoir tank	Holds water used for process
4	Rotameter	Controls the flow of water into and out of the system
5	Main-path pipe	Connects to parallel branches
6	Parallel branches	Nine pipes that consist of copper tubing, plastic pipe, and stainless steel pipe
7	Quick-connect water plugs	Connects water to the panel pipes
8	Shut-off valves	Shuts off main water supply
9	Flow-control valves	Controls flow volume of water in pipes
10	Differential pressure transmitter	Connected to pressure taps
11	Pressure taps (sensing port)	Allows determination of the appropriate pressure drops across the straight pipe sections, fittings, and orifice flowmeter
12	Thermocouple	Measures temperature of water
13	Digital readout meter	Displays temperature collected from thermocouple
14	Panel power switch	Controls power to panel
15	Pressure gauges	Determines pressure drops over various sections of the flow system

2.2 Materials and Supplies

The following table summarizes the material needed for this experiment.

Table 2: Material List with Descriptions

Index	Material/Supply Name	Description
1	Dry Compressed Air	Composition of mostly nitrogen and oxygen to be separated in the process
2	Permeable Polymeric Film	Allows for oxygen to pass through but not nitrogen
3	Oxygen Analyzers	Measure percentage of oxygen

2.3 Experimental Plan

Five experiments that are performed:

1. Rotameter Calibration
 - Perform set of experiments to determine that the conversion factor has not changed
2. Orifice Meter
 - Experimentally determine flow coefficient for the orifice flowmeter
 - Measured variables: differential pressure drop across orifice plate, volumetric flow rate, and water temperature

3. Pipe Friction Head Loss
 - Determine friction factors as a function of Reynolds number
 - Measured variables: differential pressure drop over the straight length of piping, volumetric flow rate, and water temperature
4. Pipe Fitting Head Loss
 - Determine the loss coefficient for the pipe fittings as a function of Reynold's number
 - Measured variables: differential pressure, volumetric flow rate, and water temperature

Pressure Drop Measurements:

1. Connect the pressure taps to either the DPT or two manual pressure gauges
2. Upstream pressure sensing port connects to high side of DPT and downstream pressure sensing ports connects to low side of DPT
3. Pressure transmitters should be connected to digital meters

Switching from one measurement to another:

1. Close valve to stop water flow
2. Pull out disconnect fittings from sensing ports
3. Connect fittings into sensing ports on next section measured
4. Open valve to start flow
5. Open two black valves on the side of DPT to flush out DPT
6. Close the valves on DPT
7. Wait for steady flow to read pressure drop from DPT

Project Safety Evaluation

Project Title: Fluid Flow in Pipe Systems

Date: 11/14/19

	<i>Yes/No</i>	
Potential electrical hazards?	Yes	If yes, identify conditions: Electrical outlet, wires, and digital readouts powered by electricity – near water – use caution
Potential mechanical hazards?	Yes	If yes, identify conditions: Moving valves could pinch, careful with pump, slip on water
Potential pressure hazards?	No	If yes, identify conditions:
Potential temperature hazards?	No	If yes, identify conditions:
Hazardous/toxic chemicals involved?	No	If yes, hazards involved:
Gloves required?	No	If yes, specify type:
MSDS reviewed by all team members?	Yes	List MSDS sheets reviewed: Water
<i>Process Parameters</i>	<i>Max Expected</i>	<i>List location and possible hazards and precautions</i>
Temperature (°C)	ambient	none
Pressure (psig)	20?	Pump induces some higher pressure areas, use caution
<i>Safety Controls</i>	<i>yes/no</i>	<i>If yes, Provide description</i>
	yes	Correct grounding to pipes of any electrical equipment

I have read relevant background material for the Unit Operations Laboratory entitled: “Fluid Flow in Pipe Systems” and understand the hazards associated with conducting this experiment. I have planned out my experimental work in accordance to standards and acceptable safety practices and will conduct all of my experimental work in a careful and safe manner. I will also be aware of my surroundings, my group members, and other lab students, and will look out for their safety as well.

Jake Proctor 11/14/2019
 signature of team member date

Courteney Fabbri 11/14/2019
 signature of team member date

Milos Markovic 11/14/2019
 signature of team member date

Erica Caruso 11/14/2019
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